

Nitender Kumar

Enterprise Engineering Executive | Head of Platform Architecture | Platform FinOps Governance & GenAI Orchestration
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Core Executive Profile

Engineering executive with **13+ years** building and scaling enterprise-grade platform ecosystems that convert distributed systems, data, and AI into measurable business velocity. Proven operator at the intersection of **platform engineering org design, FinOps governance, and production GenAI architecture**—orchestrating cross-functional engineering execution (Backend, Data, AI, Security) to deliver multi-tenant SaaS platforms, lakehouse analytics at **5TB+/day**, and regulated AI workloads with **sub-second decision latency, 30% cloud cost reduction, and 40% faster release cycles**.

Recognized for shifting organizations from bespoke delivery to **centralized, high-performance product development hubs**: standardizing reusable platform IP, establishing architecture governance and zero-trust controls, and driving systemic engineering velocity across global product, risk, and operations stakeholders. Deep authority across **event-driven serverless topologies, open lakehouse formats, LangGraph multi-agent orchestration, and manager-of-managers platform leadership**—positioned to own **Head of Platform, Director of Engineering, or VP Engineering** mandates where platform engineering is the enterprise-wide business accelerator.

Core Competencies Matrix

Domain	Executive Capability
Org Design & Engineering Leadership	Platform engineering org design, manager-of-managers execution, cross-functional line orchestration (Backend, Data, AI, Security), technical roadmap ownership, stakeholder alignment across India / UK / US
Platform Engineering	Multi-tenant SaaS platform architecture, reusable platform IP, CI/CD and delivery standards, self-service analytics layers, architecture governance (L1/L2/L3)
FinOps Governance	Design-time and runtime cost mapping, serverless FinOps optimization, storage and query lifecycle governance, 25–30% sustained cloud spend reduction without throughput regression
Distributed & Data Architecture	Event-driven ingestion, streaming lakehouse (Iceberg / S3), warehouse modernization (Redshift Serverless), exactly-once semantics, observability and data quality at scale
GenAI Orchestration & Security	LangGraph cyclic multi-agent systems, RAG and document intelligence pipelines, zero-trust broker patterns, Cedar RBAC/ABAC (AWS Verified Permissions), multi-tenant isolation
Engineering Velocity	Platform standardization, deployment failure reduction (50%), release predictability, downstream change-cost compression, regulated-workload acceleration
Technical Stack (Production Authority): AWS (Kinesis, Lambda, Glue Streaming ETL, S3, Iceberg, Redshift Serverless, SageMaker, Cognito, Verified Permissions), Azure (AI Document Intelligence, Cosmos DB), LangGraph, LLMs/RAG, MLOps, Hadoop/Spark lineage, CI/CD	

Leadership Highlights

- Established **platform engineering org design** and drove **systemic engineering velocity** across global engineering lines, aligning product, risk, compliance, and operations leaders on reusable roadmaps and standardized delivery patterns spanning **India, UK, and US**.
- Orchestrated engineering execution across **cross-functional engineering lines (Backend, Data, AI, Security)**, converting regulatory and business requirements into multi-tenant platform capabilities that accelerate rollout and eliminate one-off solution sprawl.
- Owned **FinOps governance** programs delivering **25–30% cloud cost reduction (~\$36K annualized)** while preserving **sub-second analytics** and production-scale throughput across **500K+ connected assets** and enterprise risk platforms.
- Institutionalized **CI/CD, MLOps, and architecture governance** enterprise-wide, cutting deployment failures by **50%** and improving release predictability by **40%**.
- Served as **principal architecture partner to C-suite and technology leadership**, defining zero-trust, cost-aware, and audit-ready platform direction for regulated, data-intensive, and AI-first enterprises.

Professional Experience

Associate Director / AI Architecture Lead — KPMG India

Dec 2025 – Present | Bangalore, India

Enterprise Context: Head of Platform Architecture for a **centralized, high-performance Product Development Hub**—building core intellectual property and scaling **multi-tenant SaaS platforms** for global enterprise tenants across risk, compliance, and architecture intelligence domains. Accountable for platform strategy, zero-trust GenAI controls, and cross-organizational engineering velocity—not billable services delivery.

TPRM — AI-Driven Enterprise Risk Platform (Multi-Tenant SaaS)

- Own end-to-end **platform architecture** for a cloud-native **Third Party Risk Management (TPRM)** product standardizing questionnaire workflows, intelligent document review, and compliance decisioning—establishing a reusable operating model that compresses manual handoffs and accelerates enterprise risk throughput.
- Architected an **event-driven integration topology** over synchronous point-to-point workflows to absorb long-running, callback-heavy risk processes—driving horizontal scale, resiliency, and end-to-end audit traceability across high-volume enterprise tenants.
- Designed **intelligent document processing pipelines** with **Azure AI Document Intelligence** for extraction, classification, and evidence indexing at scale—reducing manual review cycles and converting unstructured compliance artifacts into queryable platform assets.
- Selected **Azure Cosmos DB** for distributed workflow state to sustain callback storms, preserve append-only audit lineage, and enforce **multi-tenant isolation** without relational bottlenecks—enabling platform teams to ship risk capabilities faster with predictable scale envelopes.
- Established **architecture governance and security baselines** across L1/L2/L3 flows—data privacy, prompt-injection defense, and tenant boundary controls—creating a regulated **GenAI control plane** reusable across product lines.
- Drove **cross-functional alignment** with global risk, compliance, and operations leadership—translating regulatory mandates into standardized platform primitives that reduce bespoke design and increase enterprise rollout velocity.

Architecture Copilot Studio — GenAI Architecture Intelligence Platform

- Lead **Architecture Copilot Studio**, a shared **architecture intelligence platform** that industrializes design workflows—converting prompts and Mermaid definitions into editable **Draw.io** artifacts with persistent conversations and canvas-based editing, reducing dependency on manual architecture review cycles.
- Replaced rigid, linear **DAG prompt-chaining** with a **cyclic, stateful multi-agent orchestration layer** built on **LangGraph**, using schema-governed state machines that maintain an **append-only transaction log**—enabling recoverable agent execution, auditable reasoning traces, and controlled iteration without silent state corruption.
- Implemented a **zero-trust architecture** via a **Decoupled Security Broker Pattern**: the LLM runtime operates as an **untrusted container**, with privileged execution offloaded to an **isolated AWS Lambda Broker** inside a **private VPC**—eliminating direct data-plane access from model runtimes and shrinking blast radius for regulated workloads.
- Deployed an **Out-of-Band Auth Matrix** pairing **Amazon Cognito OIDC JWT** with **AWS Verified Permissions (Cedar policy engine)** for **RBAC** and runtime attribute cross-referencing for **ABAC**—blocking multi-tenant data leakage through policy-enforced authorization independent of model outputs.
- Introduced **Design-Time Cost Mapping** and scale-based cloud estimation (**AWS/Azure**) at architecture inception—not post-delivery—preventing **recursive agent loop token billing spikes** and enabling feasibility decisions before platform commitment.
- Published **reusable agentic reference patterns** for architecture generation, orchestration, and design optimization—giving multiple product teams a shared intelligence layer instead of fragmented, non-governed tooling.

Principal Data Engineer — Darby Telematics India Pvt Ltd & Danlaw Inc

Mar 2019 – Nov 2025 | Bangalore, India

Enterprise Context: Platform architecture owner for **Deep View Analytics**—the business-critical data and AI backbone for connected mobility, processing **5TB+/day across 500K+ connected vehicles** with sub-second operational signals for fleet health, driver risk, telematics, and emissions programs.

Deep View Analytics — Event-Driven Streaming Lakehouse (Automotive)

- Owned end-to-end platform strategy for an **AWS-native streaming lakehouse** serving as the **enterprise-wide analytics accelerator** for vehicle health, predictive maintenance, driver risk scoring, and emissions forecasting—unifying ingestion, enrichment, ML, and self-service analytics on a single governed backbone.
- Engineered a **5TB+/day event-driven AWS serverless topology**, migrating from **provisioned EMR/EC2 clusters** to an **event-driven serverless ingestion fabric** utilizing **Amazon Kinesis Data Streams**, **AWS Lambda**, **AWS Glue Streaming ETL**, and an **Apache Iceberg open table format on Amazon S3** (queried via **Amazon Redshift Serverless**)—achieving **30% FinOps cost reduction** while preserving sub-second analytics SLAs.

- Resolved **network dropout packet bursts** through a **time-windowed lookback buffer** guaranteeing **exactly-once processing semantics**—stabilizing telematics ingestion integrity across volatile connectivity profiles without duplicate or lost fleet events.
 - Integrated **AWS SageMaker pipelines** for predictive maintenance, driver risk, and emissions models—institutionalizing a repeatable **MLOps lifecycle** that moved high-value analytics from experimentation into governed production inference.
 - Delivered **GenAI-driven copilot and self-service analytics** atop the platform—democratizing natural-language access to operational and business insights and reducing engineering mediation in decision cycles.
 - Orchestrated engineering execution across **cross-functional engineering lines (Backend, Data, AI, Security)**; established **platform engineering org design, CI/CD, and data quality automation**—driving **40% faster releases** and **50% fewer deployment failures**.
 - Owned **FinOps governance** across storage tiering, Iceberg compaction strategy, and warehouse query patterns—sustaining **30% cost optimization** without compromising throughput across half-million connected endpoints.
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Senior Software Engineer — NEC India Pvt Ltd

Aug 2012 – Feb 2019 | Bangalore, India

- Architected **Hadoop-based enterprise ingestion and processing pipelines** for multi-source datasets—standardizing validation and transformation contracts to strengthen downstream reliability for analytics and IoT product lines.
 - Led delivery of a **computer vision Queue Length Management system (TensorFlow, OpenCV)** for near-real-time retail congestion detection—reducing customer wait times by **30%** and demonstrating early-scale ML operationalization.
 - Designed a **Predictive Auto Ordering System** translating demand signals into inventory recommendations—cutting stockouts by **20%** and improving revenue continuity through data-driven supply signals.
 - Shipped NEC's **IoT Big Data Solution** via modular server and distributed analytics deployments—reducing client platform setup time by **50%** and accelerating time-to-value for data-driven applications across enterprise accounts.
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Education

- **M.Tech, Data Science & Engineering** — BITS Pilani (2020–2022)
 - **B.Tech, Electronics & Communication** — GEIT, Dehradun (2007–2011)
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